## MDP & Returns

MDP $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \gamma \rangle$, dynamics $p(s', r|s, a)$.
$p(s'|s, a) = \sum_r p(s', r|s, a)$, $r(s, a) = \mathbb{E}[R_{t+1}|s, a]$
Markov: future $\perp$ past | present.
Return: $G_t = \sum_{k \geqslant 0} \gamma^k R_{t+k+1} = R_{t+1} + \gamma G_{t+1}$
Const. reward: $G_t = r/(1-\gamma)$. Episodic: $\gamma{=}1$ ok.

## Value Functions & Bellman

$v_\pi(s) = \mathbb{E}_\pi[G_t|S_t{=}s]$, $q_\pi(s, a) = \mathbb{E}_\pi[G_t|s, a]$
$v_\pi(s) = \sum_a \pi(a|s)\, q_\pi(s, a)$
Advantage: $A_\pi(s, a) = q_\pi(s, a) - v_\pi(s)$
**Bellman exp.:** $v_\pi(s) = \sum_a \pi(a|s)[r(s, a) + \gamma \sum_{s'} p(s'|s, a) v_\pi(s')]$
**Bellman opt.:** $v_*(s) = \max_a [r + \gamma \sum_{s'} p\, v_*(s')]$
$q_*(s, a) = r + \gamma \sum_{s'} p \max_{a'} q_*(s', a')$
exp. eq *linear*$\Rightarrow$closed-form ($O(|\mathcal{S}|^3)$); opt. eq *nonlinear* (max), no closed-form.
Greedy: $\pi_*(s) = \arg\max_a q_*(s, a)$ (no model; from $v_*$ needs model).

## Bandits

$Q_t(a) = \frac{1}{N_t(a)} \sum R_n \mathbb{1}_{A_n = a}$
Incremental: $Q \leftarrow Q + \alpha(R - Q)$; $\alpha{=}1/N$ avg, const. $\alpha$ recency.
Regret: $\Delta_a = q^* - q(a)$, $L_t = \sum \Delta_{A_n}$.
$\varepsilon$-greedy: random w.p. $\varepsilon$, else greedy; const $\varepsilon \Rightarrow Q \to q$ but *linear* regret; decay (GLIE)$\Rightarrow$sublinear.
UCB: $A_t = \arg\max_a [Q_t(a) + c\sqrt{\ln t / N_t(a)}]$
Thompson: sample $\tilde{q}(a) \sim P(q(a))$, act greedy.
Gradient: $\pi_t(a) = \frac{e^{H_t(a)}}{\sum_b e^{H_t(b)}}$;
$H \leftarrow H + \alpha(R_t - \bar{R}_t)(\mathbb{1}_{a=A_t} - \pi_t(a))$

## Dynamic Programming (model known)

Policy eval.: $V \leftarrow \sum_a \pi(a|s)[r + \gamma \sum_{s'} pV]$
Value iter.: $V \leftarrow \max_a [r + \gamma \sum_{s'} pV]$
Improve: $\pi'(s) = \arg\max_a q_\pi(s, a)$
Policy iter. = eval$\to$improve (finite); VI = 1 backup each.

## MC / TD / multi-step

MC: $V(S_t) \leftarrow V + \alpha(G_t - V)$ (unbiased, high var).
TD(0): $V \leftarrow V + \alpha\, \delta_t$, $\delta_t{=}R_{t+1}{+}\gamma V(S_{t+1}){-}V(S_t)$
(biased, low var, bootstraps, online).
$n$-step: $G_t^{(n)} = \sum_{l=0}^{n-1} \gamma^l R_{t+l+1} + \gamma^n V(S_{t+n})$
$\lambda$-ret.: $G_t^\lambda = (1-\lambda) \sum_n \lambda^{n-1} G_t^{(n)}$; $\lambda{=}0$ TD, $\lambda{=}1$ MC.
GAE: $\hat{A}_t = \sum_{l \geqslant 0} (\gamma\lambda)^l \delta_{t+l}$
Bootstrap: DP,TD (use estimate). Sample: MC,TD. TD both; DP neither samples.

## TD Control

SARSA (on): $Q \leftarrow Q + \alpha[R{+}\gamma Q(S', A'){-}Q]$
Exp. SARSA: target $R + \gamma \sum_{a'} \pi(a'|S') Q(S', a')$
Q-learning (off): $Q \leftarrow Q + \alpha[R{+}\gamma \max_{a'} Q(S', a'){-}Q]$
Double Q: $Q^A \leftarrow Q^A + \alpha[R + \gamma\, Q^B(S', \arg\max_a Q^A(S', a)) - Q^A]$
IS ratio: $\rho_t = \pi(A_t|S_t)/\mu(A_t|S_t)$
off-MC: $\prod \rho$; off-SARSA: $\rho_{t+1}$; Q-learn: none.

## Value Approx. & DQN

Obj.: $J(\mathbf{w}) = \mathbb{E}_\pi[(v_\pi(S) - v_\mathbf{w}(S))^2]$
SGD: $\Delta \mathbf{w} = \alpha(\text{target} - v_\mathbf{w}) \nabla_\mathbf{w} v_\mathbf{w}$
Linear: $v_\mathbf{w} = \mathbf{x}(s)^\top \mathbf{w}$; tabular = one-hot.
DQN target: $y = r + \gamma \max_{a'} Q_{\mathbf{w}^-}(s', a')$
Loss: $L = \frac{1}{B} \sum (y_j - Q_\mathbf{w}(s_j, a_j))^2$; $\mathbf{w}^-$ frozen.
Double DQN: $a^* {=} \arg\max Q_\mathbf{w}(s')$, $y{=}r{+}\gamma Q_{\mathbf{w}^-}(s', a^*)$
Dueling: $Q = V + A - \frac{1}{|\mathcal{A}|} \sum_{a'} A$
PER: $p_i \propto |\delta_i|^\alpha$, weight $w_i \propto (1/p_i)^\beta$
Noisy net: $y{=}(\mu^w{+}\sigma^w \odot \varepsilon)x + \ldots$, $\sigma$ learned.
$n$-step: $y = \sum_{k=0}^{n-1} \gamma^k R_{t+k+1} + \gamma^n \max_{a'} Q_{\mathbf{w}^-}(S_{t+n}, a')$
Distrib. (C51): $Z(s, a) \overset{d}{=} R + \gamma Z(s', a')$; 51 atoms, cross-ent. loss, act on mean.

## Policy Gradient

$J(\theta) = \mathbb{E}_{d_0}[v_{\pi_\theta}(S_0)]$ (epis.)
$= \sum_s d_\pi(s) \sum_a \pi r(s, a)$ (avg-reward).
Score trick: $\nabla \pi = \pi \nabla \log \pi$.
**PG theorem:**
$\nabla J = \mathbb{E}_\pi[\sum_t \gamma^t q_\pi(S_t, A_t) \nabla \log \pi(A_t|S_t)]$
(dynamics drop out — no model).
REINFORCE: $\theta \leftarrow \theta + \alpha \gamma^t G_t \nabla \log \pi$
Baseline $b(s)$: $\mathbb{E}[b \nabla \log \pi] = 0$;
use $G_t - v_\mathbf{w}(S_t) \approx A_\pi$ (var$\downarrow$, no bias).
Actor-critic: $\theta \leftarrow \theta + \alpha \gamma^t \delta_t \nabla \log \pi$, $\delta_t$ TD-error.
Softmax score: $\nabla \log \pi = \phi(s, a) - \mathbb{E}_\pi[\phi(s, \cdot)]$
Unified: $\nabla J = \mathbb{E}[\sum_t \Psi_t \nabla \log \pi]$, $\Psi \in \{G_t,\ G_t{-}v,\ \delta_t,\ \hat{A}^{GAE},\ q_\mathbf{w}\}$.

## Trust-Region / On-policy

Ratio $r_t(\theta) = \pi_\theta / \pi_{\theta_{old}}$.
NPG: $\theta \leftarrow \theta + \alpha F^{-1} \nabla J$, $F{=}\mathbb{E}[\nabla \log \pi \nabla \log \pi^\top]$
TRPO: $\max \mathbb{E}[r_t \hat{A}_t]$ s.t. $D_{KL}(\pi_{old} \,\|\, \pi_\theta) \leqslant \epsilon$
PPO: $L^{CLIP} = \mathbb{E}[\min(r_t \hat{A}_t,\ \text{clip}(r_t, 1{-}\varepsilon, 1{+}\varepsilon)\hat{A}_t)]$
clip active (grad=0): $r{>}1{+}\varepsilon \,\&\, A{>}0$, or $r{<}1{-}\varepsilon \,\&\, A{<}0$.

## Continuous Control

Gaussian $\pi = \mathcal{N}(\mu_\theta(s), \sigma^2)$;
score $= \frac{A_t - \mu_\theta}{\sigma^2} \nabla \mu_\theta$.
DPG: $\nabla_\theta J = \mathbb{E}[\nabla_a q_\mathbf{w}(s, a)|_{\mu_\theta} \nabla_\theta \mu_\theta]$
DDPG critic $y = r + \gamma q_{\mathbf{w}^-}(s', \mu_{\theta^-}(s'))$; actor max $q(s, \mu_\theta)$.
TD3: $y = r + \gamma \min(Q_1, Q_2)$ at $\mu_{\theta^-} + \tilde{\varepsilon}$; delayed actor.
SAC: $J = \mathbb{E}[\sum_t \gamma^t (R_{t+1} + \alpha H(\pi(\cdot|S_t)))]$, $H = -\mathbb{E}[\log \pi]$;
$y = r + \gamma\, \mathbb{E}_{a'}[\min Q_{\mathbf{w}^-} - \alpha \log \pi(a'|s')]$.

## Exploration bonuses

$r_t = r_t^{ext} + \beta\, r_t^{int}$.
Count: $r^{int} = 1/\sqrt{N(s)}$ (pseudo-counts in deep).
RND: $r^{int} = \|f_\theta(s) - f_{tgt}(s)\|^2$.
ICM: $r^{int} = \|\hat{\phi}(s_{t+1}) - \phi(s_{t+1})\|^2$.
UCB: $\mu(a) + \beta\, \sigma(a)$; ensemble var $\approx$ epistemic.

## Credit Assignment

RUDDER redistribution: $\tilde{r}_{t+1} = \hat{G}_{t+1} - \hat{G}_t$.
HER: relabel goal $g'{=}s_T$, $r'{=}\mathbb{1}_{s_t = g'}$.

## Imitation & Offline

BC: $\min_\theta \mathbb{E}\, \ell(\tilde{\pi}(s), \pi_\theta(s))$ (MSE / CE).
Compounding: $J(\pi) \leqslant J(\tilde{\pi}) + T^2 \epsilon$ (DAgger$\to$ $T\epsilon$).
DAgger roll-out: $\pi_i = \beta_i \tilde{\pi} + (1{-}\beta_i)\pi_{\theta_i}$, expert labels.
CQL: push $\downarrow$ Q(OOD) $-$ $\uparrow$ Q(data) $+ L_{TD}$ (lower bnd).
IQL: expectile $L^\tau(u) = |\tau - \mathbb{1}_{u<0}| u^2$, $u{=}Q{-}V$; $\tau{\to}1$: $V{\approx}\max_{a \in \mathcal{D}} Q$.
Decision Transf.: condition on return-to-go $\hat{R}_t = \sum_{t' \geqslant t} r_{t'}$.

## Inverse RL

Linear: $r_\psi = \psi^\top f(s, a)$, $\mu(\pi) = \mathbb{E}_\pi \sum \gamma^t f$.
Max-margin: $\min \|\psi\|^2$ s.t. $\psi^\top \mu(\tilde{\pi}) \geqslant \psi^\top \mu(\pi) + D(\tilde{\pi}, \pi)$.
MaxEnt: $p_\psi(\tau) = \frac{1}{Z_\psi} p(\tau) e^{r_\psi(\tau)}$;
$\nabla_\psi L = \mathbb{E}_{\tilde{\pi}}[\nabla r] - \mathbb{E}_{p_\psi}[\nabla r]$.
GAIL reward: $r = -\log(1 - D_\psi(s, a))$.

## Preference / RLHF / DPO

Bradley–Terry: $P(\tau_w {\succ} \tau_l) = \sigma(r_\theta(\tau_w) - r_\theta(\tau_l))$.
RM loss: $-\mathbb{E}[\log \sigma(r(y_w) - r(y_l))]$ (BCE).
RLHF: $\max_\theta \mathbb{E}_{y \sim \pi_\theta}[r_\phi(x, y) - \beta \log \frac{\pi_\theta}{\pi_{ref}}]$.
DPO reward id.: $r = \beta \log \frac{\pi_\theta}{\pi_{ref}} + \beta \log Z$;
$L_{DPO} = -\mathbb{E}[\log \sigma(\beta \log \frac{\pi_\theta(\tau_w)}{\pi_{ref}} - \beta \log \frac{\pi_\theta(\tau_l)}{\pi_{ref}})]$.
GRPO adv.: $\hat{A}_i = \frac{r_i - \text{mean}(r_{1:G})}{\text{std}(r_{1:G}) + \delta}$ (no critic).
RLVR: $r = \lambda_{fmt} r_{fmt} + \lambda_{cor} r_{cor}$.

## Planning & Search

Full tree: $b^d$ nodes.
MCTS phases: select(UCT)$\to$expand$\to$
simulate$\to$backprop; play max $N(s, a)$.
anytime; needs only a queryable sim (not differentiable); asymmetric.
UCT: $\arg\max_a [Q(s, a) + c\sqrt{\ln N(s)/N(s, a)}]$
AlphaZero $f_\theta(s){=}(\mathbf{p}, v)$; PUCT $U = Q + cP\frac{\sqrt{N}}{1+n}$;
$\pi_{MCTS} \propto N(s, a)^{1/\tau}$;
$L = (v - z)^2 - \pi_{MCTS}^\top \log \mathbf{p} + c\|\theta\|^2$.
MPC: $\arg\max \sum_k^{H-1} \gamma^k \hat{r}$, exec. first action, re-plan.
CEM: sample$\to$keep elites$\to$refit $\mu, \Sigma$.

## Model-Based RL

Model loss: $\|s' - f_\phi(s, a)\|^2$ (or $-\log p_\phi$).
Compounding rollout err $\leqslant H\epsilon$.
Ensemble var $= \underbrace{\frac{1}{M} \sum \Sigma_i}_{\text{aleatoric}} + \underbrace{\frac{1}{M} \sum (\mu_i - \bar{\mu})^2}_{\text{epistemic}}$.
Dyna-Q: real$\to$update $Q$+model; imagine $K$$\to$update $Q$. $K{=}0$: Q-learning.
Dyna-Q+: $r^+ = r + \kappa\sqrt{\tau(s, a)}$.
MBPO: SAC + short ($k{=}1$–$5$) rollouts from real states + mixed replay.
PETS: prob. ensemble + trajectory sampling (particles thru members) + MPC/CEM.
World model: $z_t{=}e_\phi(o_{\leqslant t}, a_{<t})$, predict $(z_{t+1}, r_t)$ in latent space.
World Models / Dreamer (AC in imagination) / MuZero (value-equiv. latent + MCTS).

## Taxonomy (3 axes)

**Task:** *Prediction* = policy evaluation: fixed $\pi$, estimate $v_\pi / q_\pi$ ("how good is $\pi$?"). *Control* = policy optimization: find the best $\pi$ ("what to do?"). PI/VI are control algorithms (eval+improve).
**What:** value-based ($q \to \pi$) / policy-based ($\pi_\theta$) / actor-critic (both).
**Whose data:** on-policy (eval = behavior) / off-policy ($\pi \neq \mu$, reuse data) / offline (fixed $\mathcal{D}$).
**Model:** model-based (uses $p$) / model-free.
DQN: val, off, free. REINFORCE: pol, on, free. PPO/SAC: AC. AlphaZero: AC, on, model-based.

## Bias–variance dial

MC/REINFORCE: unbiased, high var, needs full episode.
TD/AC: biased (bootstrap), low var, online.
Knobs: $n$ (steps), $\lambda$ (TD$\lambda$/GAE), $\lambda=0$ TD…$\lambda=1$ MC.
$\gamma$ = agent's discount (real); $\lambda$ = estimator var knob only.
Baseline: no bias change, big var drop $\Rightarrow$ "better/worse than typical".

## Core traps & gotchas

- $q_*$ gives $\pi_*$ *free*; $v_*$ needs the model to act.
- Continuing task needs $\gamma<1$; episodic allows $\gamma=1$.
- Q-learning needs *no* IS (max ignores $\mu$); off-MC needs product $\prod \rho$ (var explodes); SARSA needs single $\rho$.
- **Maximization bias:** $\max$ over noisy $Q$ biased *high* $\Rightarrow$ Double Q / Double DQN (decouple select vs. evaluate).
- **Deadly triad:** func. approx. + bootstrap + off-policy $\Rightarrow$ may diverge. DQN tames it (replay + target net).
- **SARSA vs Q (cliff):** SARSA = safe (learns the noisy policy it follows); Q = optimal/risky (learns greedy). Equal once $\varepsilon\to 0$.
- Semi-gradient: ignore $\nabla$ through TD target.
- $\gamma^t$ in PG usually *dropped* in practice.
- Expected SARSA: lower var than SARSA (averages $A'$).

## Stable PG / continuous

**PPO two anchors:** KL-to-$\pi_{\mathrm{ref}}$ (SFT) = where you *end up* (anti-overopt); clip-to-$\pi_{\mathrm{old}}$ = how far you *step*. Don't conflate.
PPO clip $\Rightarrow$ multiple epochs per batch.
DDPG = DQN + learned argmax $\mu_\theta$ (continuous).
**TD3 = DDPG + 3 fixes:** clipped double-Q ($\min$), delayed actor, target-policy smoothing.
**SAC:** entropy in the *objective* (not a regularizer) $\Rightarrow$ changes which policy is optimal; sustained exploration + robustness.
Use: PPO (parallel sims, LLMs), TD3 (clean cts. baseline), SAC (real robots, sample-eff.).

## Exploration

Undirected: $\varepsilon$-greedy, action noise, entropy, noisy nets — answer "how to randomize", not "where".
Directed = uncertainty / novelty / surprise.
Bootstrapped DQN + randomized priors $\Rightarrow$ deep (temporally consistent) exploration.
**Noisy-TV problem:** pure prediction-error curiosity gets stuck on unpredictable noise; can't tell "unfamiliar" from "random". Fix: NGU (episodic+lifelong), clip/anneal $\beta$.

## Credit assignment (real quiz)

3 root causes: **depth** (delay), **density** (sparse), **breadth** (many paths).
Var of MC $\uparrow$ with reward delay; TD($\lambda$) assigns credit faster than TD(0).
Tools: TD/$n$-step/GAE (mild), RUDDER (long delay), HER (sparse goals), advantage (breadth).
**SDP:** Markov policy + Markov transitions, reward *need not* be Markov.
**Return-equivalent:** same *expected return per episode.*
**Explaining-away:** later states predict return $\Rightarrow$ earlier causal actions get missed.
Zero future reward $\Rightarrow$ must enrich state w/ past $\Rightarrow$ breaks Markov.
**SA vs CA:** SA = how small input change moves output (gradient); CA = how input is *responsible* (not gradient). LSTM errors correlated $\Rightarrow$ cancel in differences (low var).
Potential-based shaping $\equiv$ $Q$-init $Q'=Q+\Phi$ (online $\Phi$ re-adds bias).
Optimality preserved by *strictly monotonic transform of total return* (not per-step affine/clip/square).

## Imitation & Offline

BC trains on $p_{\mathrm{data}}$, deploys on $p_{\pi_\theta}$ $\Rightarrow$ shift; error $T^2\epsilon$. DAgger fixes to $T\epsilon$ but needs a *queryable* expert.
BC traps: non-Markov (history), causal confusion (brake-light), multimodal (MSE averages modes $\to$ split the obstacle; fix = MoG / CVAE / autoregressive).
**Offline fails:** $\max_{a'} Q$ queries OOD actions $\Rightarrow$ overestimate $\Rightarrow$ bootstrapped blow-up.
Principle: *don't trust the model OOD.* 3 families: behavior-reg, conservative value (CQL), avoid query (IQL).
**Stitching:** Bellman methods (CQL/IQL) *can*; Decision Transformer *cannot* (imitates returns).
IL upper bound = expert; offline RL *can exceed* the dataset.

## IRL & reward learning

**IRL ill-posed:** const. $r$ makes all $\pi$ optimal $\Rightarrow$ need a principle (max-margin / max-entropy).
IRL learns the *reward* (transfers); IL just copies actions.
Start = expert demos; assume dynamics often known.
MaxEnt: keep max randomness s.t. matching expert feature stats; gradient = $\uparrow$expert − $\downarrow$policy reward.
**Bottleneck:** partition function $Z$ = sum over all trajectories (differentiate through it / fwd-bwd).
Slack var = allow suboptimal expert. $p(\tau)$ = dynamics, reward-independent.
Feature matching: match $\mu(\pi)$; needs solving an MDP each step.

## Alignment / LLMs

Pretraining = BC from text; not enough (plausible, conflicting; no instruction-follow / refusal).
**DPO** removes the reward model *and* RL loop $\to$ supervised on pairs; limit: fixed offline prefs.
**GRPO** drops the value/critic net; baseline = group-standardized rewards; $\hat{A}_i$ sequence-level, same for all tokens of $y_i$.
**RLVR:** automatic verifier (math/code/tests); composite format+correctness (else capability underrated).
SFT: demos (limited). RLHF: prefs (proxy hacking). RLVR: verifier (narrow / hack verifier).
Gato: tokenize all modalities $\Rightarrow$ one Transformer; trained by supervised sequence modeling.

## Reward hacking / Goodhart

"When a measure becomes a target it ceases to be a good measure."
Overoptimization: true perf rises then *falls*; mitigate with $D_{\mathrm{KL}}(\pi \,\|\, \pi_{\mathrm{ref}})$.
Examples: boat-race spinning for boosts; camera moved so object *looks* grasped; verbose/sycophantic RLHF.

## Planning & Model-based

MCTS plays **highest visit count**, not highest $Q$. MCTS = policy-improvement operator; net amortizes the search.
AlphaZero vs AlphaGo: drops human games, rollout policy, supervised pretrain; one net $(p, v)$.
MPC executes *first* action only, then re-plans $\Rightarrow$ corrects model error.
Random shooting (uniform, keep best) vs CEM (refit distribution to elites).
Dyna-Q $K=0$ $\Rightarrow$ plain Q-learning.
**Aleatoric** = env noise (irreducible); **epistemic** = lack of data (reducible, matters for planning). Single Gaussian net = aleatoric only; need an *ensemble* for epistemic.
Model traps: dist. shift, compounding error, *model exploitation* (policy finds glitches).
Fix: short $H$, mix real data, penalize uncertainty.
MBRL helps: data expensive, smooth dynamics, short $H$, dense reward.
**Latent world model (Dreamer/MuZero):** raw obs often wrong for control (high-dim, partial, distractors) $\Rightarrow$ learn $z$ useful for *control*, not reconstruction. PETS *trajectory sampling* = roll particles thru ensemble $\Rightarrow$ carries uncertainty across the horizon.

## Quick families

DQN line: Double, PER, Dueling, multi-step, distributional (C51), noisy $\to$ **Rainbow** (all 6).
Value: TD, SARSA, Q, DQN. Policy: REINFORCE, TRPO, PPO. AC: A2C/A3C, DDPG, TD3, SAC. Model-based: Dyna-Q, MPC, AlphaZero, MuZero, Dreamer.

## Convergence conditions

**GLIE:** every $(s,a)$ visited $\infty$ often *and* policy $\to$ greedy; e.g. $\varepsilon_t=1/t$ ($\varepsilon_t\to 0$, $\sum \varepsilon_t=\infty$).
**Robbins–Monro** steps: $\sum_t \alpha_t=\infty$, $\sum_t \alpha_t^2<\infty$ (e.g. $\alpha_t=1/t$).
Tabular Q-learning $\to$ $q_*$ if all $(s,a)$ seen $\infty$ often + RM steps (any behavior).
Greedy bandit: linear regret; $\varepsilon$-decay / UCB / Thompson: sublinear.

## Numbers worth knowing

$\mathbb{E}[\max \text{ of } 3\ \mathcal{N}(0,1)] \approx 0.85$ (max-bias).
PPO clip $\varepsilon \approx 0.1$–$0.2$; GAE $\lambda \approx 0.9$–$0.97$.
$n$-step sweet spot $3$–$5$; MBPO rollout $k=1$–$5$.
DQN: $84\times 84$ gray, 4 stacked frames, 49 Atari, human-level on 29.
Rainbow $\approx 7\times$ fewer steps; MBPO $\approx 20\times$ more sample-eff.
Effective horizon $\approx \frac{1}{1-\gamma}$ ($\gamma=0.99 \to 100$ steps; $\gamma\to 0$ myopic).

## Why stochastic / POMDP

Value-based greedy $\Rightarrow$ *deterministic* policy.
Need stochastic when: (i) optimum is mixed (rock–paper–scissors $\to \frac{1}{3}$ each); (ii) aliasing / POMDP (aliased gridworld $\to$ coin-flip escapes).
Policy-based gives both for free; value-based cannot.

## State representation (SRL)

Good latent: structure, continuity, sparsity/disentangle.
Methods: bisimulation (reward+dynamics metric), auxiliary (recon/fwd/inv-dyn/reward), data augmentation, contrastive (NCE), non-contrastive (BYOL), attention.
Reconstruction wastes capacity on salient-but-irrelevant pixels.
End-to-end (aligned, data-hungry) vs SRL+RL (general, maybe misaligned).